

TABELA DE DERIVADAS DE FUNÇÕES REAIS
Cursos: Engenharias

Disciplina: Análise Matemática I

Docentes: Grupo de disciplina

Nível: I

Semestre: 2º/2024

Carga Horária: 6h/Semana

Considere:

- u e v como funções deriváveis de x ;
- a e n como constantes reais tais que $a > 0$ e $n \neq 0$.

Nº	Função	Derivada	Função Composta	Derivada da Composta
1	$y = x^n$	$y' = nx^{n-1}$	$y = u^n$	$y' = nu^{n-1} \cdot u'$
2	$y = \ln x$	$y' = \frac{1}{x}$	$y = \ln u$	$y' = \frac{u'}{u}$
3	$y = a^x$	$y' = a^x \ln a$	$u = a^u$	$y' = u' \cdot a^u \cdot \ln a$
4	$y = \log_a x$	$y' = \frac{1}{x \ln a}$	$y = \log_a u$	$y' = \frac{u'}{u \ln a}$
5	$y = e^x$	$y' = e^x$	$y = e^u$	$y' = u' \cdot e^u$
6	$y = \sin x$	$y' = \cos x$	$y = \sin u$	$y' = u' \cos u$
7	$y = \cos x$	$y' = -\sin x$	$y = \cos u$	$y' = -u' \sin u$
8	$y = \tan x$	$y' = \sec^2 x$	$y = \tan u$	$y' = u' \cdot \sec^2 u$
9	$y = \cot x$	$y' = -\operatorname{cosec}^2 x$	$y = \cot u$	$y' = -u' \cdot \operatorname{cosec}^2 u$
10	$y = \sec x$	$y' = \sec x \tan x$	$y = \sec u$	$y' = u' \cdot \sec u \tan u$
11	$y = \operatorname{cosec} x$	$y' = -\operatorname{cosec} x \cot x$	$y = \operatorname{cosec} u$	$y' = -u' \cdot \operatorname{cosec} u \cot u$
12	$y = \arcsin x$	$y' = \frac{1}{\sqrt{1-x^2}}$	$y = \arcsin u$	$y' = \frac{u'}{\sqrt{1-u^2}}$
13	$y = \arccos x$	$y' = -\frac{1}{\sqrt{1-x^2}}$	$y = \arccos u$	$y' = -\frac{u'}{\sqrt{1-u^2}}$
14	$y = \arctan x$	$y' = \frac{1}{1+x^2}$	$y = \arctan u$	$y' = \frac{u'}{1+u^2}$
15	$y = \sinh x$	$y' = \cosh x$	$y = \sinh u$	$y' = u' \cdot \cosh u$
16	$y = \cosh x$	$y' = \sinh x$	$y = \cosh u$	$y' = u' \cdot \sinh u$
17	$y = \tanh x$	$y' = \frac{1}{\cosh^2 x}$	$y = \tanh u$	$y' = \frac{u'}{\cosh^2 u}$
18	$y = \coth x$	$y' = -\frac{1}{\sinh^2 x}$	$y = \coth u$	$y' = -\frac{u'}{\sinh^2 u}$
19	$y = \operatorname{arcsenh} x$	$y' = \frac{1}{\sqrt{1+x^2}}$	$y = \operatorname{arcsenh} u$	$y' = \frac{u'}{\sqrt{1+u^2}}$
20	$y = \operatorname{arcosh} x$	$y' = \frac{1}{\sqrt{x^2-1}}$	$y = \operatorname{arcosh} u$	$y' = \frac{u'}{\sqrt{u^2-1}}$
21	$y = \operatorname{arctanh} x$	$y' = \frac{1}{1-x^2}$	$y = \operatorname{arctanh} u$	$y' = \frac{u'}{1-u^2}$
22	$y = x^x$	$y' = x^x (1 + \ln x)$	$y = u^v$	$y' = u^v \ln u \cdot v' + v \cdot u^{v-1} \cdot u'$

Nº	Fórmulas Trigonométricas	
1	$\tan \alpha = \frac{\sin \alpha}{\cos \alpha}$	$\cot \alpha = \frac{\cos \alpha}{\sin \alpha}$
2	$\sec \alpha = \frac{1}{\cos \alpha}$	$\operatorname{cosec} \alpha = \frac{1}{\sin \alpha}$
3	$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \sin \beta \cos \alpha$	$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \sin \beta \cos \alpha$
4	$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$	$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$
5	$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$	$\sin \alpha - \sin \beta = 2 \sin \frac{\alpha - \beta}{2} \cos \frac{\alpha + \beta}{2}$
6	$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$	$\cos \alpha - \cos \beta = -2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$
7	$1 + \cos 2\alpha = 2 \cos^2 \alpha$	$1 - \cos 2\alpha = 2 \sin^2 \alpha$
8	$\sin(2\alpha) = 2 \sin \alpha \cos \alpha$	$\cos(2\alpha) = \cos^2 \alpha - \sin^2 \alpha$
9	$1 + \tan^2 \alpha = \sec^2 \alpha$	$1 + \cot^2 \alpha = \operatorname{cosec}^2 \alpha$
10	$\sin^2 \alpha + \cos^2 \alpha = 1$	$\tan \alpha \pm \tan \beta = \frac{\sin(\alpha \pm \beta)}{\cos \alpha \cos \beta}$